# Assignment II

## Part A

**Muhammed Shameel KV**
**bmat2330@isibang.ac.in**

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1. Determine (with justification) whether the following statements are true or false:

Every continuous function can be uniformly approximated by a sequence of even polynomials.

**Answer:** The statement is true since even polynomials form an algebra. We can see this as follows: if are 2 even polynomials then,

Hence even polynomials on form an algebra

Since even polynomials contain the constant function (the polynomial ) and separate points on (for any , , the polynomial is even and separates and ), by the **Stone-Weierstrass theorem**, the set of even polynomials is dense in . Thus, every continuous function on can be uniformly approximated by a sequence of even polynomials.

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Every continuous function can be uniformly approximated by a sequence of odd polynomials.

**Answer:** This statement is also true. If is any continuous function, we can consider a function . Now this is also continuous since in . Hence by we have that can be uniformly approximated by a sequence of even polynomials . Now consider the sequence of odd polynomials where . such that, for

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Every continuous function can be uniformly approximated by a sequence of even polynomials.

**Answer:** This statement is also true. The same argument as in goes through.

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Every continuous function can be uniformly approximated by a sequence of odd polynomials.

**Answer:** This statement is false since all odd polynomials needs to be at . Hence any sequence of odd polynomials should converge pointwise at to i.e. . So continuous functions with cannot be uniformly approximated.

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2. Fix some function . Define a sequence of functions by

If forms an equicontinuous family then show that is constant.

**Answer:** By assumption of equicontinuity we have, such that,

Since such that . So we get for ,

Here is arbitrary so we have that,

and since is also arbitrary, we have . Hence is a constant function.

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3. Let be a sequence of continuous functions such that

Define a sequence of functions by

Show that has a subsequence that converges uniformly on .

**Answer:** We have that

We see that for any we have such that for we have , for all . Hence is uniformly equicontinuous. Now,

Hence is uniformly bounded, and is compact. So by **Arzela-Ascoli theorem** we have that has a convergent subsequence which converges uniformly on .

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4. Let be a continuously differentiable function on . Show that there is a sequence of polynomials such that

**Answer:** We have that is continuous on so by **Weierstrass approximation theorem** we have that such that uniformly. Now define,

We see that since uniformly implies, such that for all and . Hence for we have,

Hence we have that converges uniformly to . Also we have by construction of that . So by **Fundamental theorem of calculus** we have,

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5. Give an example (with justification) of a continuous function which cannot be approximated by any sequence of polynomials on .

**Answer:** Every non-constant polynomial is unbounded. Hence if we choose any non-constant bounded continuous function we have such that

Hence we have that is bounded for , i.e. is constant for . Hence it can only converge to a constant function. So any non-constant bounded continuous function cannot be approximated by polynomials on . For example, is a continuous bounded function which cannot be approximated by polynomials on .